Multi-Agent Power Market Simulation using EMCAS

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***Abstract*—Countries around the world continue to restructure their electricity markets and open them up to competition and private investors in pursuit of economic efficiency and new capital investment. However, the recent volatility exhibited by many restructured power markets, in combination with several prominent market failures, have highlighted the need for a better understanding of the complex interactions between the various market participants and the emerging overall market behavior. Advanced modeling approaches are needed that simulate the behavior of electricity markets over time and model how market participants may act and react to changes in the underlying economic, financial, and regulatory environments. This is particularly useful for developing sound market rules that will allow these markets to function properly. A new and promising approach is to model electricity markets as complex adaptive systems using an agent-based modeling and simulation approach, such as is implemented in the Electricity Market Complex Adaptive System (EMCAS) software. EMCAS provides an agent-based framework to capture and investigate the complex interactions between the physical infrastructures and the economic behavior of market participants that are a trademark of the newly emerging markets. This paper describes the EMCAS agents, their interactions, the unique insights obtained from agent-based models, and discusses current model applications in several U.S., Asian, and European markets.**

***Index Terms*—regional electricity market, interconnections, power transactions, locational marginal prices.**

I. Introduction

The complex interactions and interdependencies among participants in today's deregulated, decentralized electricity markets are much like those studied in game theory (Picker, 1997). However, the strategies used by many power market participants are often too complex to be conveniently modeled by standard game theoretic techniques. In particular, the ability of market participants to repeatedly probe markets and rapidly adapt their strategies adds additional complexity. Computational social science offers appealing extensions to traditional game theory.

Computational social science involves the use of agent-based modeling and simulation (ABMS) to study complex social systems (Epstein and Axtell, 1996). ABMS consists of a set of agents and a framework for simulating their decisions and interactions. ABMS is related to a variety of other simulation techniques, including discrete event simulation and distributed artificial intelligence or multi-agent systems (Law and Kelton, 2000; Pritsker, 1986). ABMS tools are designed to simulate the interactions of large numbers of individuals so as to study the macro-scale consequences of these interactions (Tesfatsion, 2002). Each entity in the system under investigation is represented by an agent in the model. An agent is thus a software representation of a decision-making unit. Agents are self-directed objects with specific traits and typically exhibit bounded rationality, that is, they make decisions by using limited internal decision rules that depend only on imperfect local information. In practice, each agent has only partial knowledge of other agents and each agent makes its own decisions based on the partial knowledge about other agents in the system.

Several electricity market ABMS tools have been constructed, including those created by Bower and Bunn (2000), Petrov and Sheblé (2000), Lai et al. (2000), Skoulidas et al. (2002), Veselka et al. (2002), and North et al. (2002). These models have demonstrated the potential of agent simulations to act as electronic laboratories, or "e-laboratories," suitable for repeated experimentation under controlled conditions.

II. Modeling Approach

While planning and operation of traditional electric utility systems used to be strongly driven by least-cost and reliability concerns, recent trends toward restructuring and unbundling are creating opportunities for new participants with new business models to enter the markets and are creating diverse and dynamic markets. Centralized, monopolistic decision-making organizations are giving way to heterogeneous, decentralized decision structures. The "single" decision-maker is replaced by a host of decision entities each with their own, unique business strategies, risk preferences, and decision models.

The implicit assumption of a centralized decision-making process built into many of the global optimization and equilib-

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rium-based power systems analysis tools developed over the last two decades limits their ability to adequately analyze the forces prevalent in today's emerging markets. This is amplified by recent experiences that have shown the difficulties of understanding the operation of these markets.

Unlike conventional electric systems analysis tools, the Electricity Market Complex Adaptive System (EMCAS) model does not postulate a single decision maker with a single objective for the entire system. Rather, agents are allowed to establish their own objectives and apply their own decision rules. The complex adaptive systems modeling approach simulates agents that learn from their previous experiences and change their behavior when future opportunities arise. That is, as the simulation progresses, agents can adapt their strategies on the basis of the success or failure of previous efforts. Genetic algorithms are used to provide a learning capability for certain agents. With its agent-based approach, EMCAS is specifically designed to analyze multi-agent markets and allow testing of regulatory structures before they are applied to real systems.

III. Model Agents

The model includes a large number of different agents to capture the heterogeneity of restructured markets (see Fig. 1), including generation companies (GenCos), transmission companies (TransCos), distribution companies (DistCos), independent system operators (ISOs) or regional transmission organizations (RTOs), demand companies (DemCos), consumers, and regulators. Agents are highly specialized to perform diverse tasks ranging from acting as generation companies to modeling transmission lines. To support specialization, agents include large numbers of highly specific rules. Analysts can easily define new rules and strategies to be used by the agents and then examine the marketplace consequences of these strategies.

Decisions on how and when to operate its equipment and what prices to charge for its output are made separately by each agent based on its business strategy. This business strategy, however, does not have to remain static. Rather, agents may change strategies as learning and adaptation occurs. In the model, agents learn about market behavior and the actions of other agents using two forms of learning, that is, *observation-based learning* and *exploration-based learning*. Under the observation-based learning, agents evaluate their past performance and decide to either (1) maintain the current business strategy, (2) adjust the current business strategy, or (3) switch to a new business strategy. Using exploration-based learning, agents explore various marketing and bidding strategies and observe the results of their actions. Once a strategy is found that performs well, it is exercised and fine-tuned as subtle changes occur in the marketplace. When more dramatic market changes take place and a strategy begins to fail, an agent will more frequently explore new strategies in an attempt to adapt to the dynamic and evolving supply and demand forces in the marketplace. Even when a strategy continues to perform well, agents periodically explore and evaluate other strategies in their search for one that performs better. Through this process, agents engage in a price discovery process and learn how they may potentially influence the market through their own actions to increase their corporate utility. A company that has learned that it has little influence over the market or is risk-averse may decide not to increase bid prices.

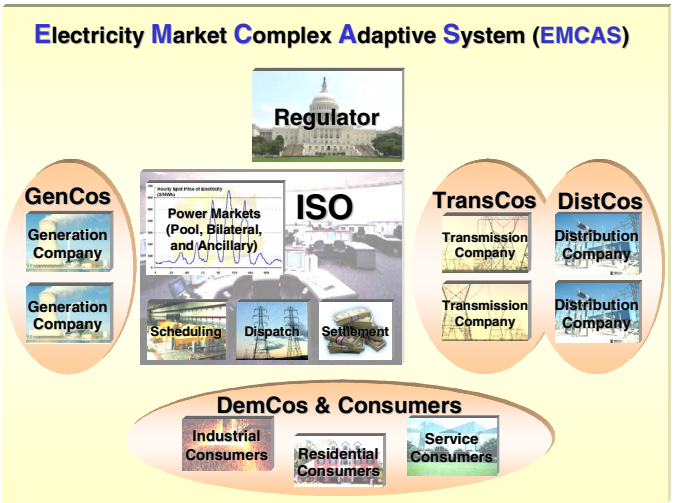

Fig. 1: Model Agents

The business profiles of agents are described using multi-attribute utility theory. A business profile consists of objectives, risk preference, and a utility function. The business objectives are the parameters that the company uses to determine the desirability of the various options it has available. The most commonly used business objective is profit. Other objectives can be included. The agent's risk preference is determined by a parameter that characterizes the company as risk-neutral, risk-prone, or risk-averse. The objectives and risk preference are combined to form a company utility function. Each agent seeks to maximize its expected utility throughout the simulation, and each agent can have a different business profile to test the effects of different company business styles.

IV. Interaction Layers

The interaction layers provide the environment for the agents to operate in. This environment is typically multi-dimensional, that is, agents operate within several interconnected layers, including a physical layer, several business layers, and a regulatory layer as shown in Fig. 2.

The physical layer at the bottom of the figure represents the agents that are involved in the physical generation, transmission, distribution, and consumption of electricity. Consumers, generators, and transmission nodes and links together make up the physical part of the electricity market. The ISO/RTO operates both the transmission system and the electricity markets. In the physical layer, the ISO/RTO exercises its dispatch function to operate the system to match load and generation and to adjust to changes in load, generator or transmission outages, and other unplanned events.

Fig. 2 also shows three business layers that represent the business side of the electricity market. Here, the GenCo agents that own the generators make the decisions regarding how to operate the equipment to meet company objectives. The

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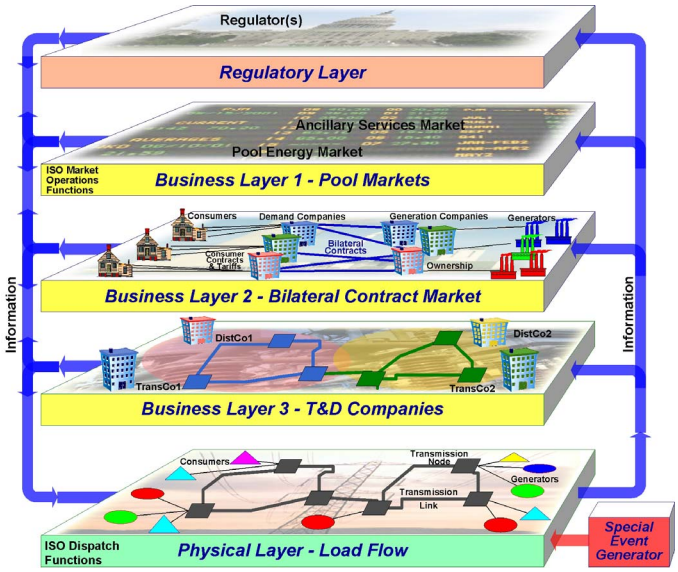

Fig. 2: Interaction Layers

DemCos participate in the electricity market by buying electricity from GenCos and selling it to their consumers. The model can account for situations where a GenCo and a DemCo are part of the same corporate parent. Pool markets for energy and ancillary services serve as central clearing houses for buyers and sellers and are operated by the ISO/RTO agent.

GenCos and DemCos can engage in bilateral contract markets for the sale and purchase of electricity. These contracts are negotiated privately between two agents. The transmission and distribution company layer is designed to account for the ownership of the transmission and distribution systems and for the fees charged by these companies for the use of their facilities. The TransCos and DistCos may be part of a single corporate parent, along with a GenCo and DemCo. The model can account for this corporate connection while maintaining a separate accounting of each business unit.

The regulator is the agent in the regulatory layer that sets the market rules and monitors market performance. Currently, the user provides input as the regulator.

V. Model Markets

Currently, the model can simulate three types of markets that include bilateral contract, pool energy, and ancillary services. Generally, bilateral contracts are agreements between a single GenCo agent and a single DemCo agent. These contracts have time scales that range from hours to several years. The model simulates bilateral contracts among generation company and demand agents through a series of requests for proposals (RFPs) that are initiated by the DemCo agents.

In the pool energy market, agents submit buy and sell bids to the auction clearinghouse—in this case, the ISO. There are two pool market options. The first is the locational marginal price option in which all agents get paid the marginal bid to serve loads at a specific location (uniform auction). The second pool market option is referred to as "Pay-as-Bid" in which each GenCo gets paid the price that it bids (discriminatory auction). In the day-ahead pool market, GenCos' bidding strategies are formulated for the entire day, not for individual hours. GenCos use public information as well as private information to formulate their bidding strategies.

The model simulates three ancillary services markets after the pool market closes. These include regulation, spinning reserve, and replacement reserve markets. The amount of these services that is purchased by the ISO is a function of system reliability and security parameters that are entered by the user.

VI. Long-Term System Expansion

On the multi-year planning level, agents engage in capacity expansion planning. This section presents the capacity expansion algorithm for GenCo agents, though with certain modifications, it applies to other agents as well. The approach attempts to model decentralized expansion decisions by diverse market participants with unique behaviors. Construction decisions, announcements, and schedules are based on rational company-level behavior to achieve certain corporate objectives.

Multi-attribute utility functions determine the best course of action for each independent market player. In evaluating whether to build a new power plant, for example, GenCo agents review and rank all available generation options according to their agent-specific market positions, risk profiles, and business objectives. A new merchant company agent with limited access to capital markets will value the various generation technologies differently than a large, established company agent. Some agents will focus on small, peaker-type plants, while others will also consider technologies more suitable for large, base load-type generation.

Once started, project construction schedules are fluid; that is, agents continuously adapt to market conditions. At the beginning of each simulation year, GenCos review their capacity expansion plan in light of current and projected market conditions and make a go/no-go decision on adding new generating capacity to their portfolio. GenCo agents may decide to accelerate a project if prevailing market conditions appear more favorable than initially projected, or delay, put on-hold, or cancel projects entirely if the market outlook changes for the worse.

Agents use the process outlined in Fig. 3 to reach their expansion decisions. Agents develop demand expectations as well as market price projections to estimate future profits and evaluate the financial viability of adding new capacity. Agents take into account expected price volatility that is partly due to planned maintenance outages as well as random forced outages. Future forced outages are based on Monte-Carlo simulations. Projected prices are expressed as price probabilities over all random draws. Agents use this information to construct price exceedance curves (Fig. 4) for each month over several years to account for prices and price volatility over the entire range of demands. Agents use the price exceedance curves to estimate the potential capacity factors and expected profits of new facilities for each month over several years.

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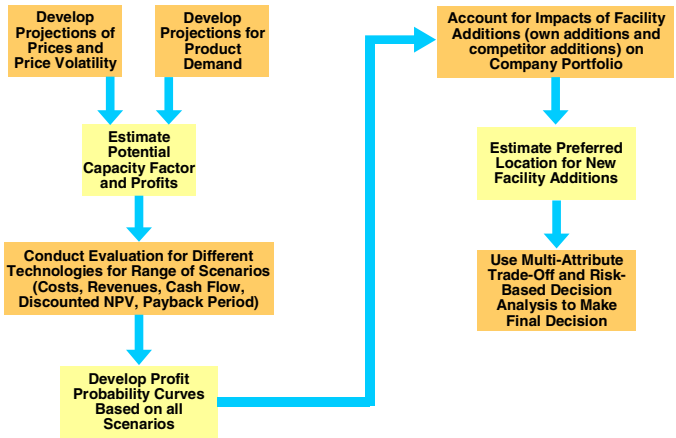

Fig. 3: Expansion Decision Flow Chart

Using this information, agents estimate financial parameters, such as net present value, discounted cash flow, and project pay-back period. Agents may use these parameters as criteria in their decision-making process to screen out least profitable technologies from further consideration. To accommodate uncertainties, company agents evaluate technology profitability under several possible future outcomes using a scenario tree.

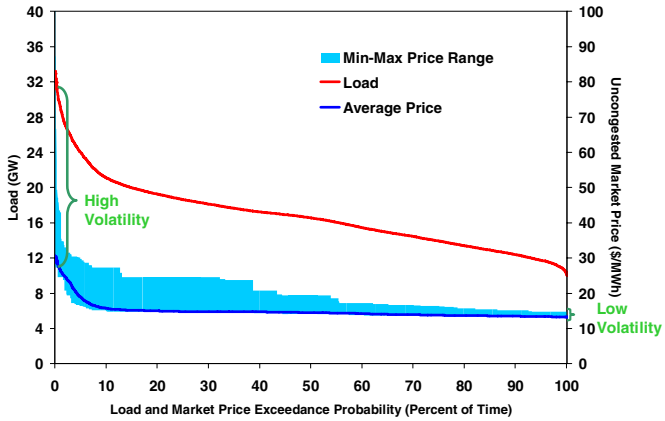

Fig. 4: Price Exceedance Curves

The profitability of each technology under all outcomes is plotted in profit exceeedance curves. Some technologies may be shown to be unprofitable over the entire range of scenarios, while even the most promising technologies may turn out to lose money under certain conditions. Profit exceedance curves are designed to quickly illustrate this by showing the probability x that a given technology makes a profit of at least y. In the example shown in Fig. 5, there is a 20% probability that Technology 3 will make a profit of $2.23 billion (NPV) or more over its economic lifetime, and a 65% probability for a profit of at least $425 million. The curves also show the probabilities of a technology breaking even, as well as the maximum payoff and the maximum loss or risk exposure associated with each option. While on average the two technologies in Fig. 5 look fairly similar, the curves show a different picture in terms of upside benefits and downside risks. While a gambler may choose to build Technology 3 with the higher maximum payoff, a risk-averse agent may select Technology 2 to limit the downside potential.

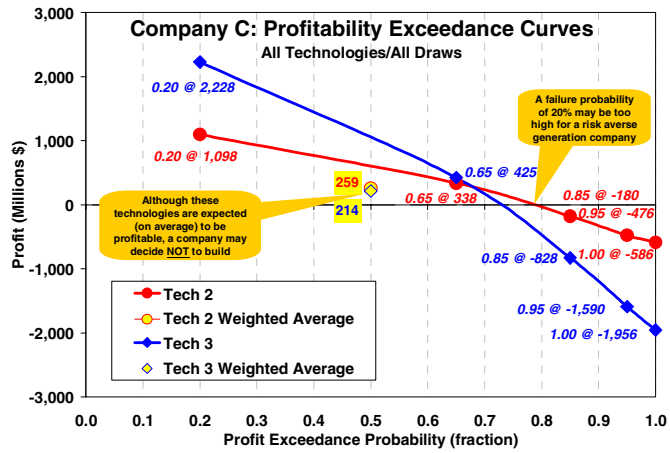

Fig. 5: Technology Profit Exceedance Curve

Agents can take into account expansion decisions by their competitors. All expansion plans and facility additions are made known to all agents. A company agent can account for these additions when constructing monthly market price exceedance curves, though it does not know with certainty if these plans will come into fruition due to cancellations, build-schedule acceleration, build-schedule delays, or false signals to competitors. New unit construction uncertainty can be accounted for as a set of scenarios and associated probability of occurrence.

The approach also accounts for situations when the utilization and profitability of a GenCo's existing units are decreased by adding a new unit to its portfolio. An energy networking algorithm is used to identify demand pockets and potential locations for adding new facilities. Agents make their final build decision after going through a decision analysis process. This process accommodates each agent's unique risk profiles and specific objectives. If agents have more than one business objective, this step provides an opportunity to analyze trade-offs across different criteria. Agents select a specific technology based on the total corporate utility computed in this process.

VII. Recent Model Applications

The model was recently applied to study the future power market configuration in a part of the Midwestern United States. The system under consideration included about 240 generators, about 2,000 buses, over 30 GenCos, and multiple TransCos and DistCos.

Results show that throughout the year, there is sufficient generation to meet the study-area load. Results also show that in low-load hours it is possible to use the least-cost generation since transmission congestion does not occur. However, in intermediate-load hours and, even more pronounced, in peak-load hours, some of the lower-cost units must be bypassed and more expensive units need to be dispatched out of economic merit. This congestion-related change in dispatch leads to LMP differences across the system (Fig. 6 shows an example of this situation for a hypothetical region).

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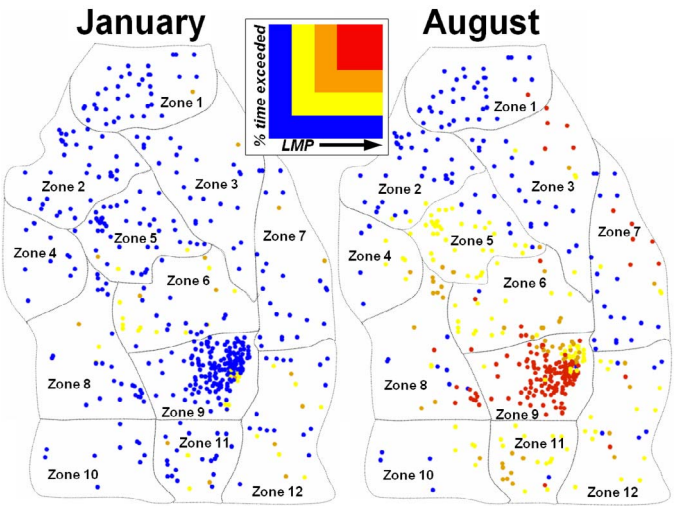

Fig. 6: Potential Future Load Pockets

Additional model runs tried to identify the potential for companies to exploit their strategic position to profitably raise prices above marginal production cost. The intent was to determine if the market structure as currently envisioned would be able to function efficiently and effectively given the physical limitations of the power system and the competitive forces of the marketplace. Business strategies that a company might pursue in order to improve its position include physical withholding (single or multiple units), economic withholding, and various price probing strategies. Model runs show that the ability to exercise market power varies substantially across the different GenCos, with some companies able to drive up prices substantially and others with no impact at all.

Companywide economic withholding during all 24 hours of a peak-load day is attractive to some GenCos but not to others. Fig. 7 shows the impact on one generation company if it were to price all of its units in its portfolio increasingly above production cost. Results are shown for the peak day only. Portfolio generation drops as bid prices increase above production cost. The reduction in output initially leads to a decrease in operating profits for the company up to about 150% above cost as less expensive generation and adequate transmission capacity is available to meet the load, both from study-area and out-of-area sources. However, as bid prices continue to increase (200% above production cost), operating profits start

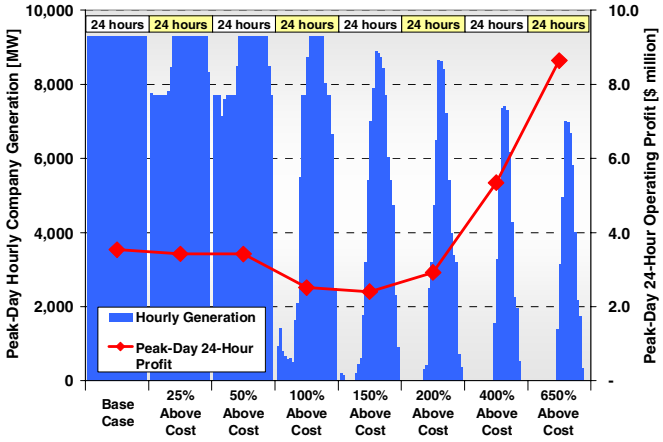

Fig. 7: GenCo1 24-Hour Strategic Bidding

to grow rapidly as a certain portion of the company's capacity is needed to meet the load and transmission constraints limit the use of cheaper capacity from elsewhere.

Fig. 8 compares generation and operating profits for the same company over a range of bid prices for the above strategy (bid prices increased for all 24 hours) with a strategy where units are priced higher only during the afternoon from 2 to 6 p.m., when the load is the greatest. While daily output falls substantially if GenCo1 were to apply its pricing strategy for the entire day, production remains fairly unchanged should the company try to exert its market power only in the afternoon. This appears to be a more attractive strategy for GenCo1 over most of the pricing range, particularly if the company is concerned

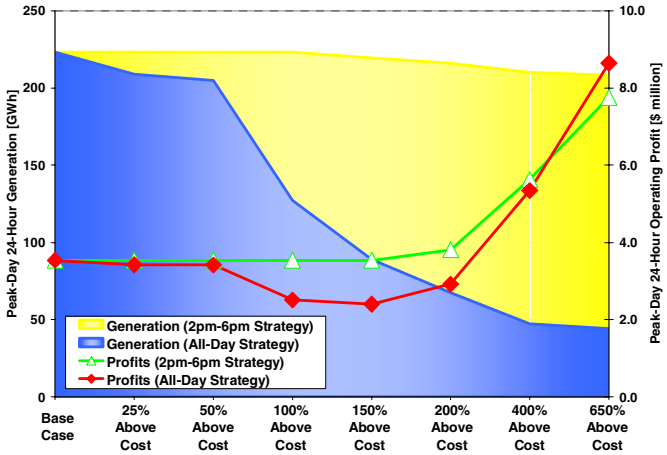

Fig. 8: GenCo1 Results of Different Strategies

with excessive cycling of its units as generation under this strategy shows little variation throughout the day even as prices are increased significantly.

Fig. 9 shows results for a GenCo that is unable to exert market power. As GenCo2 increases its bid price during the entire day (24 hours) for its portfolio of units, output rapidly declines and finally reaches zero as all of its generators will eventually be replaced by cheaper units. Operating profits turn negative, that is, GenCo2 actually starts losing money under this strategy as it will continue to incur fixed operating expenses. Even if the company applies this portfolio strategy only during the afternoon peak hours, it still will face reduced profitability as some

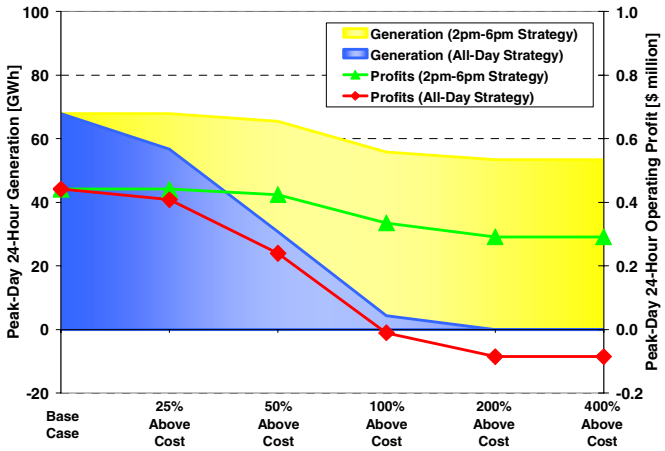

Fig. 9: GenCo2 Results of Different Strategies

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of its units drop out of the dispatch and the price increase does not compensate for the lower portfolio capacity factor.

The model is currently applied to study restructuring issues in South Korea (Fig. 10) and in Western Europe (Fig. 11). Under a new project, the model will also be applied to analyze the Iberian Power Market.

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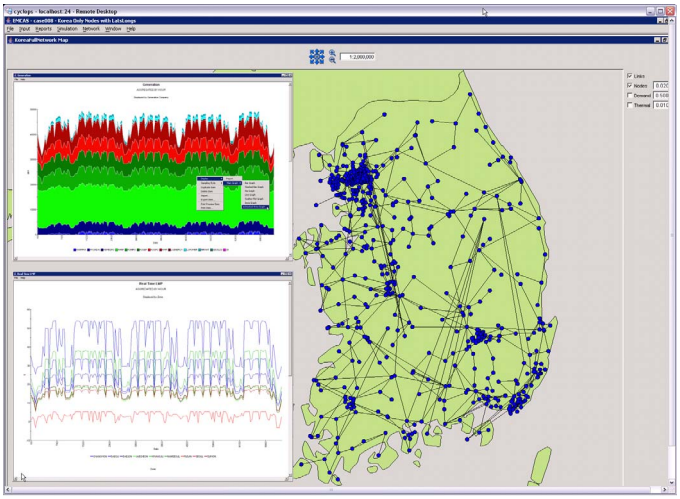

Fig. 10: Model Application in South Korea

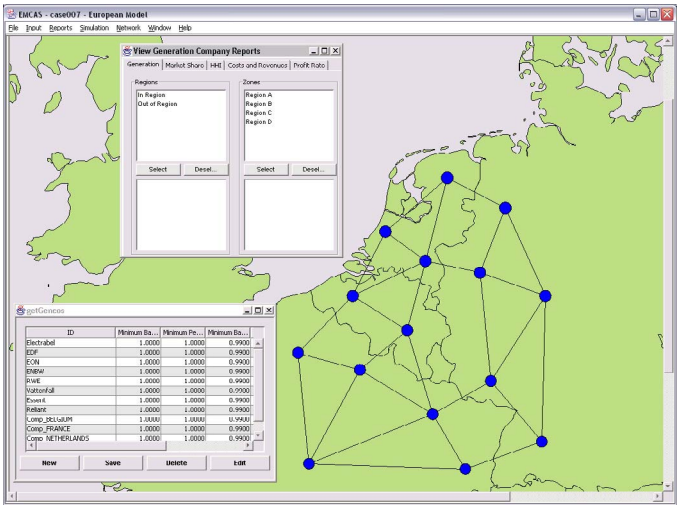

Fig. 11: Model Application in Western Europe

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IX. Biographies

Guenter Conzelmann is the Deputy Director of the Center for Energy, Environmental, and Economic Systems Analysis (CEEESA) at Argonne National Laboratory. Mr. Conzelmann has an environmental engineering degree from the University of Bingen in Germany.

Gale Boyd has been at Argonne for 20 years. He received a Ph.D. in Economics from Southern Illinois University. His research activities include modeling industrial energy demand, emissions, and productivity.

Vladimir Koritarov graduated in 1982 from the School of Electrical Engineering, University of Belgrade, Yugoslavia. In 1991 he joined Argonne National Laboratory, U.S.A., where he is presently an Energy Systems Engineer in CEEESA.

Tom Veselka is an Energy Systems Engineer in CEEESA. He has provided technical support and managed projects related to electric utility systems and the environment for over 20 years.

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